Levels of polybrominated diphenyl ethers (PBDEs) and organochlorine pollutants in two species of Antarctic fish (Chionodraco hamatus and Trematomus...

In the late 1960s the first scientific studies on contamination in Antarctica demonstrated the presence of pollutants in Antarctic ecosystems. Many Persistent Organic Pollutants (POPs) are transported globally from the areas in which they are produced and released into the environment in remote areas, including Antarctica. Here we report results obtained concerning the accumulation of polybrominated diphenyl ethers (PBDEs), mono- and non-ortho-polychlorobiphenyls (PCBs), polychlorodibenzodioxins (PCDDs) and polychlorodibenzofurans (PCDFs) in the tissues of two species of Antarctic fish (Chionodraco hamatus and Trematomus bernacchii). The 2,3,7,8-TCDD toxic equivalents (TEQs) were also calculated to evaluate the potential risk of these compounds for the two species. In general, POP levels were higher in the tissues of T. bernacchii than in C. hamatus and the highest concentrations were found in the liver of both species. The PBDE levels varied from 160.5 pg g(-1) wet wt in C. hamatus muscle to 789.9 pg g(-1) wet wt in T. bernacchii liver and were lower than the levels of PCBs. PCBs were the main organochlorine compounds detected and their concentrations ranged from 0.3 ng g(-1) wet wt in C. hamatus muscle to 15.1 ng g(-1) wet wt in T. bernacchii liver. TEQ concentrations resulted higher in C. hamatus than in T. bernacchii and were due mainly to PCDDs. The presence of PBDEs and organochlorine pollutants in the tissues of Antarctic organisms confirms their global transport and distribution.